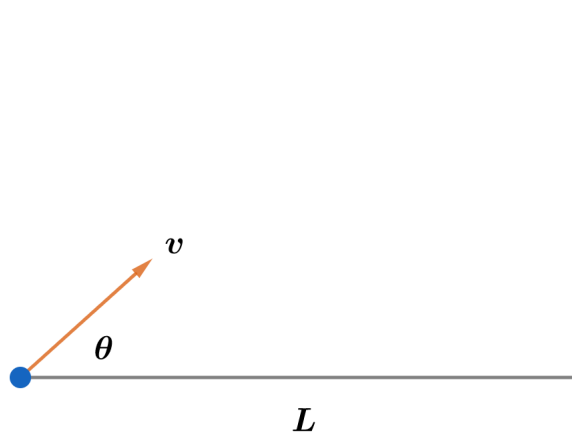


2019B F=ma Exam: Problem 14

Kevin S. Huang



Recall the range equation from projectile motion

$$R = \frac{v^2 \sin(2\theta)}{g}$$

The wall serves to reflect the ball's trajectory in the horizontal direction so

$$2L = \frac{v^2 \sin(2\theta)}{g}$$

$$v = \frac{\sqrt{2Lg}}{\sqrt{\sin(2\theta)}}$$

and the answer is E.